

CS & IT ENGINEERING



Computer Network

Supernetting

Lecture No. - 06



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Recap of Previous Lecture



Topic

Subnetting





Topics to be Covered



Topic

Supernetting



[MCQ] [GATE-2019] [2-Mark]

#Q. Consider three machines M, N and P with IP addresses "100 . 10 . 5 . 2", "100 . 10 . 5 . 5" and "100 . 10 . 5 . 6" respectively. The subnet mask is set to "255 . 255 . 252" for all the three machines. Which one of the following is true?

- ☒ A M, N and P all belong to the same subnet
- ☐ B Only N and P belong to the same subnet
- ☐ C Only M and N belong to the same subnet
- ☐ D M, N, and P belong to three different subnets

Ans: B

Subnet Mask :

255.255.255.252

=

255.255.255.11111100

30 bits Prefix (NetID, subnetID) → 2 bits hostID

M : 100.10.5.2

=

100.10.5.00000010

N : 100.10.5.5

=

100.10.5.000000101

P : 100.10.5.6

=

100.10.5.000000110



Topic : Subnet Size



- Maximum number of IP addresses that can be assigned to a network interfaces
- Maximum number of hosts can be in a sub-network



Topic : Subnet Size



→ HostID field (y_2 bits)

Sub-network Size = $[2^{y_2} - 2]$ hosts per subnet



Topic : Number of subnets

→ Subnet ID field (y_1 bits)

Number of subnets = $[2^{y_1}]$ subnets per network

→ Subnet ID all zero and all one are allowed. (Default)

Example 5 :-

[MCQ]



#Q. Which one of the following CIDR prefixes exactly represents the range of IP addresses 80.81.82.0 to 80.81.82.255?

☒ A 80.81.0.0 / 16

☒ B 80.81.82.0 / 24

☐ C 80.81.82.0 / 25

☐ D 80.81.82.0 / 26

Ans: B

80.81.82.0
80.81.82.255

: 80.81.82.00000000 } (2^8 Address)
: 80.81.82.11111111

CIDR Prefix

: 80.81.82.00000000 /24
prefix 24 bits 8 bit Host ID

80.81.82.0 /24



Topic : Supernetting



→ Prefix Aggregation / CIDR Aggregation / Route Aggregation



Topic : Supernetting



→ Combining (logically) smaller networks into single large network

Example 6 :-

#Q. Consider following Network Addresses of networks. What should be the supernet address ?

Network Address₁ : 50 . 60 . 70 . 0 / 26

Network Address₂ : 50 . 60 . 70 . 64 / 26

Network Address₃ : 50 . 60 . 70 . 128 / 26

Network Address₄ : 50 . 60 . 70 . 192 / 26

→ 2^6 Add.

→ 2^6 Add.

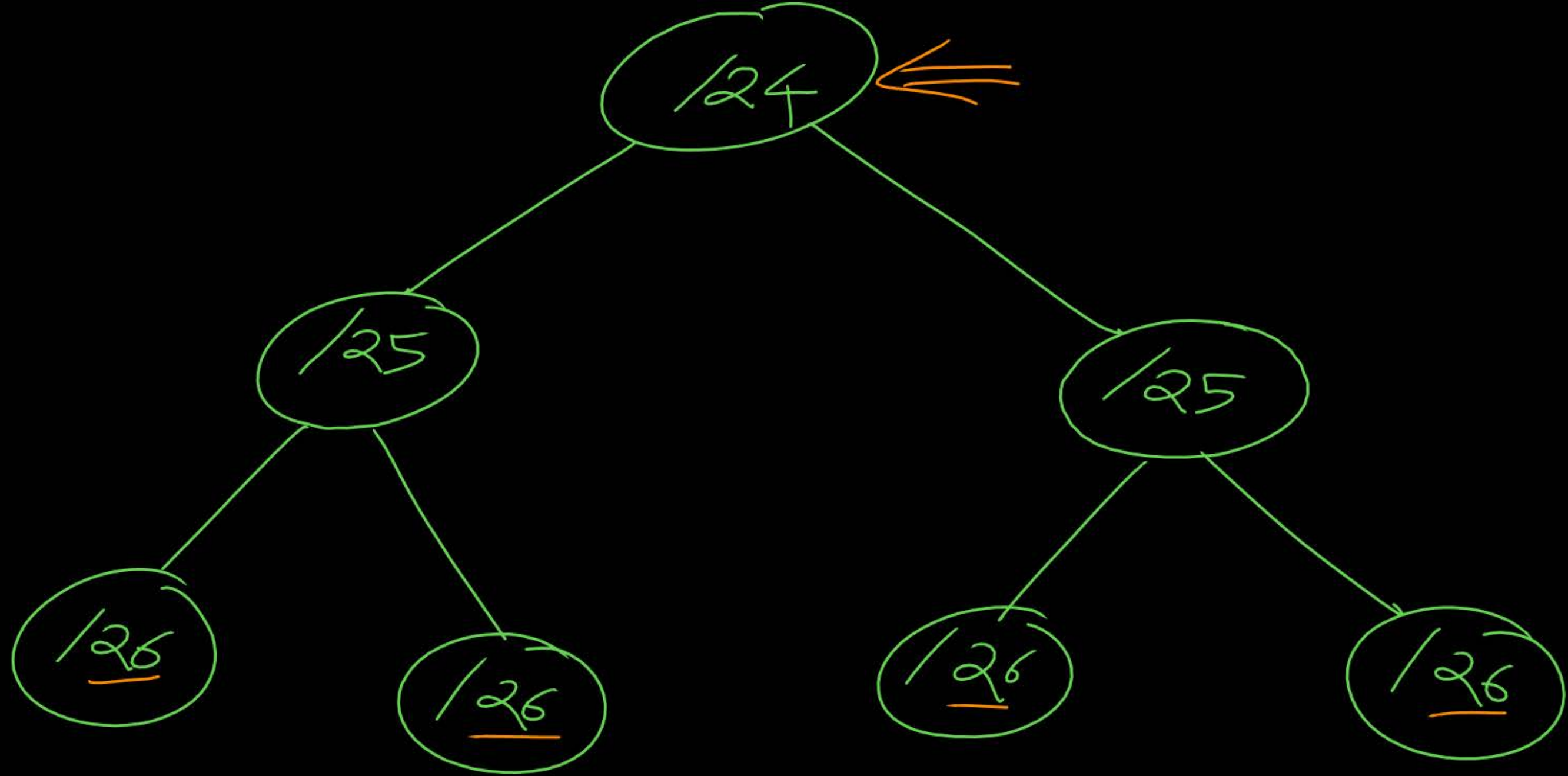
→ 2^6 Add.

→ 2^6 Add.

(12) → 2^7 Add.
(125)

(34) → 2^7 Add.
(125)

(1234) → 2^8 Add.
(125)



Solution 6 :-

$$50.60.70.0 / 26 \quad : \quad 50.60.70.00000000 / 26$$

$$50.60.70.64 / 26 \quad : \quad 50.60.70.01000000 / 26$$

$$50.60.70.128 / 26 \quad : \quad 50.60.70.10000000 / 26$$

$$50.60.70.192 / 26 \quad : \quad 50.60.70.11000000 / 26$$

26 bit prefix 6 bit Host ID

50.60.70.0 / 26 : 50.60.70.0 00000000 / 26

50.60.70.64 / 26 : 50.60.70.0 10000000 / 26

6 bit
Host ID

1-bit supernet

Supernet Address₁₂

: 50.60.70.0 00000000 / 25

25 bit prefix 7 bit
Host ID

$50.60.70.128 / 26$: $50.60.70.10000000 / 26$
 $50.60.70.192 / 26$: $50.60.70.11000000 / 26$

A double-headed vertical arrow indicates the difference in the 4th octet between the two addresses, specifically between the 10 and 11 bits.

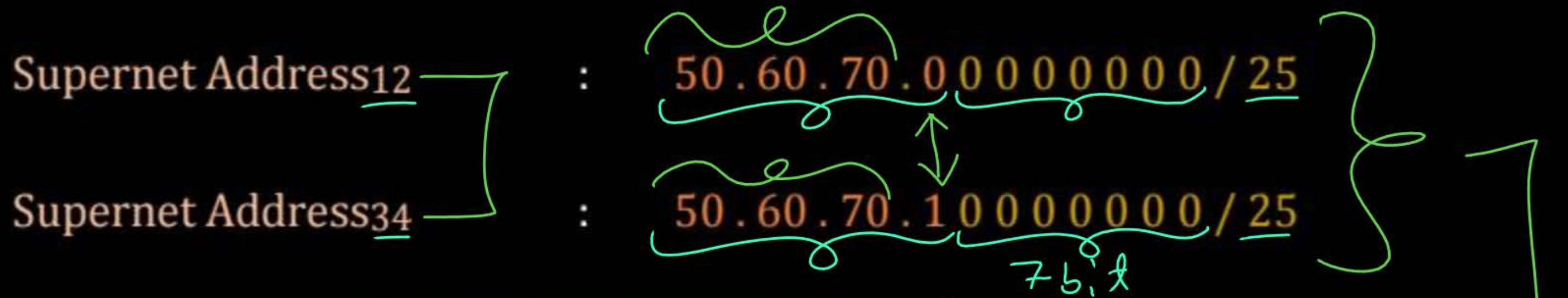
Supernet Address₃₄ : $50.60.70.10000000 / 25$

25 bit prefix 7 bit Host ID

Supernet Address₁₂ : 50.60.70.00000000 / 25

Supernet Address₃₄ : 50.60.70.10000000 / 25

7 bit



1-bit supernet

Supernet Address₁₂₃₄ : 50.60.70.00000000 / 24

24 bit Prefix 8 bit HostID



$50.60.70.0 / 26$: $50.60.70.00000000 / 26$
 $50.60.70.64 / 26$: $50.60.70.01000000 / 26$
 $50.60.70.128 / 26$: $50.60.70.10000000 / 26$
 $50.60.70.192 / 26$: $50.60.70.11000000 / 26$

2-bit supernet

Supernet Address1234 : $50.60.70.00000000 / 24$
 8bit host ID
 Ans: $50.60.70.0 / 24$
 24 bit prefix

Network Address 1 : 50 . 60 . 70 . 0 / 26

First IP = 50 . 60 . 70 . 0 / 26, Last IP = 50 . 60 . 70 . 63 / 26

Network Address 2 : 50 . 60 . 70 . 64 / 26

First IP = 50 . 60 . 70 . 64 / 26, Last IP = 50 . 60 . 70 . 127 / 26

Network Address 3 : 50 . 60 . 70 . 128 / 26

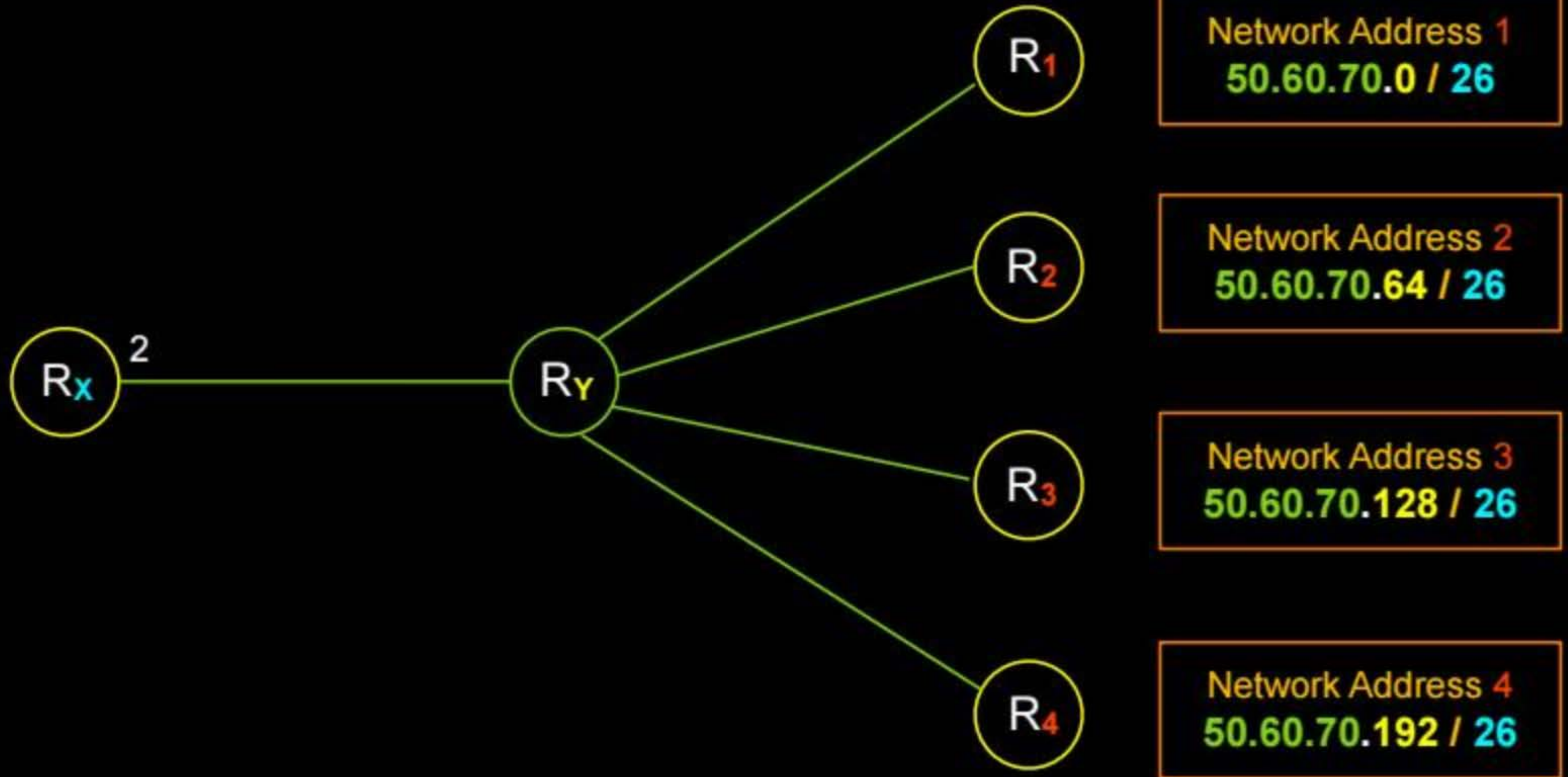
First IP = 50 . 60 . 70 . 128 / 26, Last IP = 50 . 60 . 70 . 191 / 26

Network Address 4 : 50 . 60 . 70 . 192 / 26

First IP = 50 . 60 . 70 . 192 / 26, Last IP = 50 . 60 . 70 . 255 / 26



Topic : Supernetting





Router (R_x) Forwarding (Routing) table

Network Address	Interface ID	Next Hop
50.60.70.0 / <u>26</u>	2	R _y
50.60.70.64 / <u>26</u>	2	R _y
50.60.70.128 / <u>26</u>	2	R _y
50.60.70.192 / <u>26</u>	2	R _y



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⇒ Optimization of Routing Table



Router (Rx) Forwarding (Routing) table
After Route (Prefix / CIDR) Aggregation

Network Address	Interface ID	Next Hop
<u>50.60.70.0 / 24</u>	2	Ry

↓
2-bit
Supernet Add.



Topic : Supernetting



→ Reduces the number of entries in routing table

* This may lead less
processing overhead at router



Topic : Supernetting



→ Allow more efficient routing → fast Routing



Topic : Supernetting

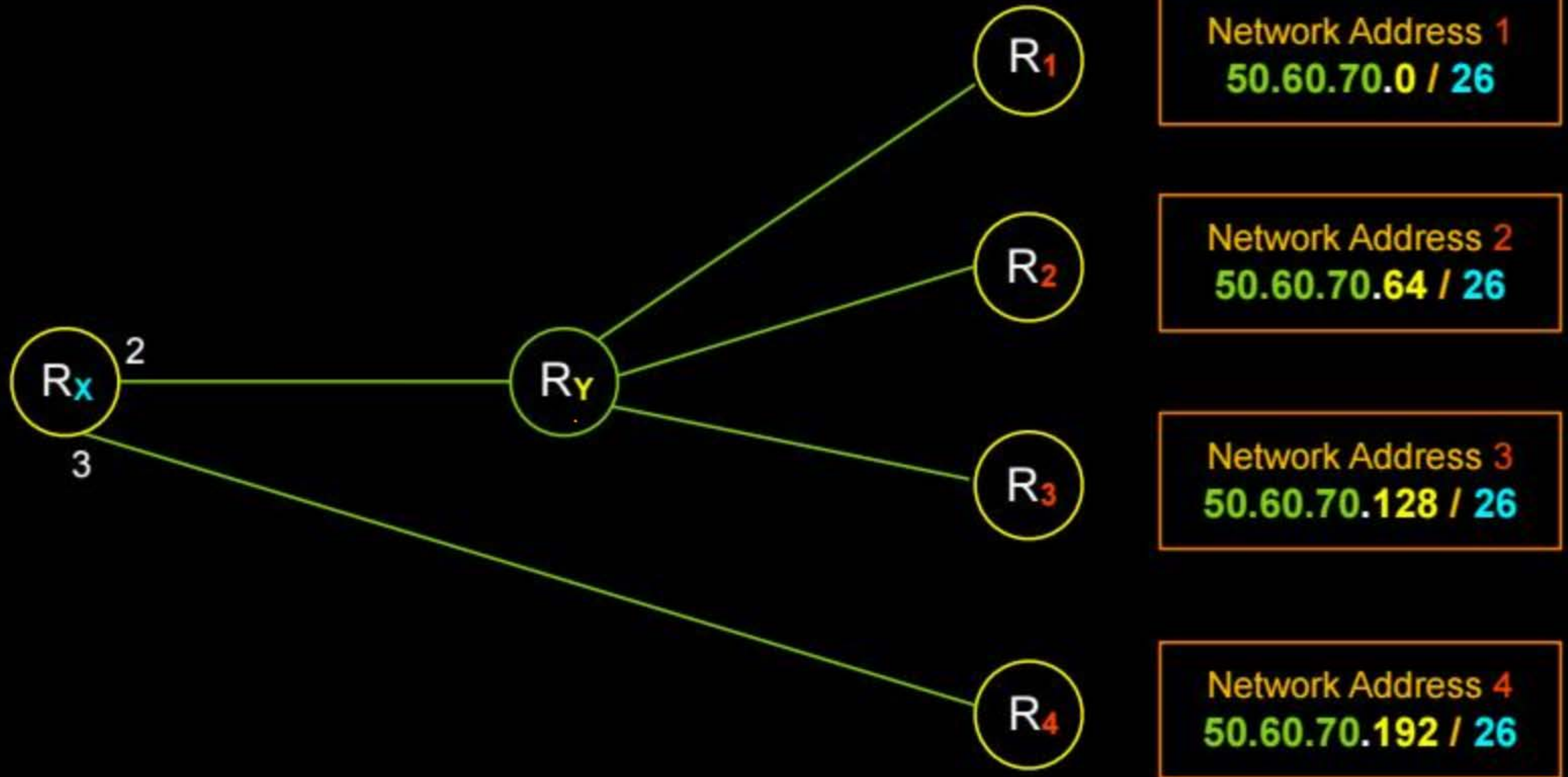


→ All the **Networks** should be **contiguous**

[Block of addresses having **contiguous prefixes**]



Topic : Supernetting





Topic : Supernetting



Router (R_x) Forwarding (Routing) table

Network Address	Interface ID	Next Hop
50.60.70.0 / 26	2	R _Y
50.60.70.64 / 26	2	R _Y
50.60.70.128 / 26	2	R _Y
50.60.70.192 / 26	3	→ R ₄



Topic : Supernetting



Router (R_x) Forwarding (Routing) table
After Route (Prefix / CIDR) Aggregation

Network Address	Interface ID	Next Hop
<u>50.60.70.0 / 25</u>	2	R _y
<u>50.60.70.128 / 26</u>	2	R _y
<u>50.60.70.192 / 26</u>	3	R ₄



2 mins Summary



Topic

Supernetting

① What *

② How $\Rightarrow$

③ Why *

(S-22) $\rightarrow$ IIT
KGP

(S-23) $\rightarrow$ IIT-K
CIDR
Prefix



THANK - YOU